

HARSHA MELUR NAGESH

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SUMMARY

Software Developer with 2+ years of experience specializing in AI-driven backend systems and Agentic AI solutions using Python, Java, and cloud technologies. Skilled in building autonomous multi-agent systems using LangGraph, CrewAI, AutoGen, and MCP, delivering intelligent workflows with real business impact. Proven track record - reduced API latency by 30%, improved model accuracy from 45% to 95%, and boosted application performance by 43%. AWS Cloud Practitioner certified, with a strong foundation in full-stack development, CI/CD, and enterprise automation, driven to build scalable AI solutions.

EDUCATION

Illinois Institute of Technology, Chicago, Illinois August 2024 - May 2026
Master of Science, Information Technology and Management

Visvesvaraya Technological University, Bengaluru, India. August 2018 - May 2022
Bachelor of Engineering, Computer Science

SKILLS

Frontend: React.js, Vue.js, Angular, TypeScript, JavaScript, HTML5, CSS3, Bootstrap.

Backend: Java (Spring Boot), Python (FastAPI), Node.js, Express.js, REST APIs, Hibernate/JPA.

Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLite.

Gen AI/ Agentic AI: OpenAI Agents SDK, LangGraph, CrewAI, AutoGen, MCP, n8n, LangChain, RAG, FAISS, Ollama, OpenAI API, Claude API.

Cloud & DevOps: AWS (EC2, S3), Redshift, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Terraform.

System Design & Security: Data structures and Algorithms, System Design, Security (OAuth, JWT), Operating Systems (Windows, MacOS, Linux, Red Hat, Ubuntu), Computer Networks (TCP/IP, ARP, UDP, HTTP/HTTPS), Multithreading and Concurrency, API Design.

Soft Skills: Problem-Solving, Adaptability, Time Management, Critical Thinking, Teamwork, Communication, Collaboration, Leadership.

WORK EXPERIENCE

SOFTWARE DEVELOPER

- Iruve, Bengaluru, India** May 2022 - June 2024
- Designed and implemented scalable RESTful backend services using Python (FastAPI) to support product catalog, cart, and order workflows, leveraging ACID-compliant database transactions and isolation levels to ensure data consistency under concurrent traffic.
 - Reduced API response latency by 30% by optimizing SQL queries, indexing strategies, and introducing a Redis caching layer, improving system throughput and checkout reliability under simulated high-load conditions.
 - Implemented retry logic, structured exception handling, and failure recovery mechanisms to maintain transactional integrity and ensure resilient order processing.
 - Conducted load testing simulating 1,000+ concurrent users, analyzing throughput and latency metrics, and integrated structured logging and monitoring to improve observability and production stability.
 - Developed responsive frontend components using Vue.js, and deployed containerized services using Docker on AWS, automating builds and deployments through CI/CD pipelines within an Agile SDLC environment.
 - Built and maintained API-driven workflows and automation processes to streamline internal operations and improve system interoperability across services.
 - Led a team of 5 in image annotation, dataset preparation, model training, and testing for an internal computer-vision project, ensuring high-quality labeled data and consistent evaluation processes.
 - Annotated and curated datasets using roboflow, contributing to a model accuracy improvement from 45% to 95% through better data quality and structured workflows.
 - Trained and evaluated models using YOLOv8, optimizing preprocessing and training steps to achieve a 30% reduction in processing time.
 - Attended client meetings to present progress updates, clarify requirements, and align technical deliverables with business expectations.

PROJECTS

AI Engineering Team Agent: (CrewAI, LangGraph, Docker, Python)

- Deployed a 4-agent autonomous system to manage, build, and test software applications using CrewAI and Coder Agents in Docker, demonstrating multi-agent orchestration and real-world agentic AI workflows.

Autonomous Trading Floor Agent (MCP, CrewAI, AutoGen, Python)

- Built a multi-agent trading platform with 4 autonomous agents powered by 6 MCP servers and 44 tools, enabling real-time autonomous decision-making and trade execution.

Full-Stack Web Application: SafariX (J2EE, Servlets, JSP, JPA/Hibernate, MySQL, Payara, JUnit, Maven, REST APIs, MVC Architecture)

- Built a full-stack wildlife safari booking system using J2EE and MVC Architecture with CRUD operations, optimized database interactions, and integrated MCP server to enable AI-assisted features using Claude AI for intelligent interactions.

CERTIFICATION

- AWS Cloud Practitioner (CLF-C02).
- Introduction to Model Context Protocol by Anthropic.
- Model Context Protocol Advanced Topics by Anthropic.